# DOCGUARD FORENSIC AUDIT CERTIFICATE

**CERTIFICATE ID:** DG-EA365BB0-A588-49
**ISSUED:** 2026-10-30 20:02:39 UTC

Automated Multimodal Document Tampering & Cryptographic Integrity Verification

*Standard: ISO/IEC 27037 & FRE 902(14)*

## FORENSIC VERDICT: TAMPERED / FRAUD DETECTED

The analyzed document exhibits conclusive empirical indicators of digital manipulation (Global Forgery Risk Score: 98.46%). Visual ELA discontinuities, non-balancing arithmetic, or structural PDF revision anomalies were confirmed.

FORGERY RISK SCORE
**98.5%**
Threshold: 50.0%

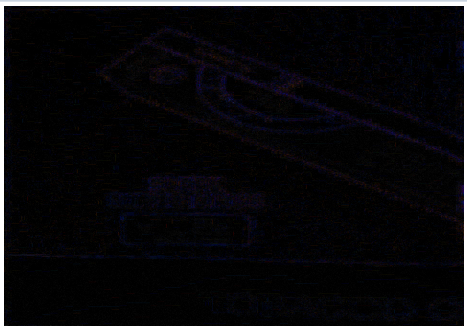
## 1. Legal Chain of Custody & Document Identity

| | | | |
|---|---|---|---|
| **Original Filename:** | **WhatsApp Image 2026-10-31 at 1.26.33 AM.jpeg** | **File Size:** | 193.48 KB (198,126 bytes) |
| **Analysis Time:** | 2026-10-31 01:32:39 (Local) | **Audit System:** | DocGuard Automated Forensic System v1.0 |
| **SHA-256 Checksum:** | **DA06DD95A86284E6BD0B151CACC1ECFD198423B179BA58BEF4CA32D016830DF6** | **MD5 Checksum:** | **60387812493F8A4215D0ADE0B9B47C6A** |

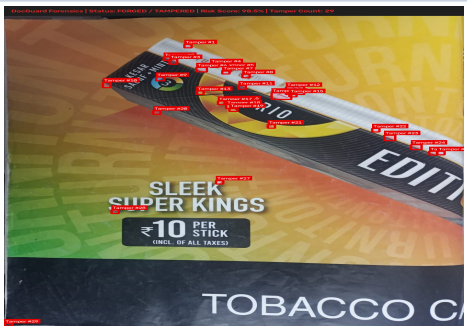
## 2. Tri-Panel Visual Forensic Evidence




**Panel A: Raw RGB Document**
Normalized Document Canvas



**Panel B: In-Memory ELA Map**
JPEG Compression Residuals




**Panel C: XAI Anomaly Map**
Bounding Boxes & Translucent Overlay

## 3. Six-Factor Forensic Scorecard Breakdown

| Forensic Vector / Factor | Methodology / Analysis Engine | Observed Metric / Diagnostic | Safety Threshold | Status |
|---|---|---|---|---|
| **1. Deep Multimodal Fusion Brain** | Late Fusion (ResNet-50 + EfficientNet-B0) | 98.5% Probability | < 50.0% Risk | **FAIL** |
| **2. Error Level Analysis (ELA)** | Differential JPEG Q=95 Quantization | High-Variance Seams Detected (Δ > 15x) | Uniform Residuals | **FAIL** |
| **3. Spatial Noise Residuals** | 3x3 Median Spatial Residual Variance | Localized Sensor Noise Attenuation | Homogeneous Poisson Noise | **ALERT** |
| **4. PDF Binary Architecture** | Linearized %%EOF Revision Audit | Multiple Trailer Revisions (Count >= 2) | Single Trailer (Count == 1) | **FAIL** |
| **5. Font Seams & Typography** | Embedded Type1/TrueType Font Audit | Discrepant Font Glyph Encodings | Consistent Kerning & Fonts | **ALERT** |
| **6. Deterministic Arithmetic Ledger** | Symbolic Rule Verifier (sum(Items) == Total) | Non-Balancing Ledger Discrepancy | Discrepancy == 0.00 | **FAIL** |

## 4. Symbolic Arithmetic Ledger Consistency Audit

*Note: Document is a non-structured graphical image without native PDF text streams. Ledger verification fell back to high-frequency visual edge and ELA neural stream checks.*

## 5. Forensic Adjudication Notes & Findings

Automated multi-modal inspection conducted by DocGuard v1.0 on 2026-10-30 20:02:39 UTC. The primary neural fusion classifier assigned a 98.46% probability of tampering. Critical tampering confirmed: Localized JPEG DCT quantization anomalies and non-balancing arithmetic amounts establish deliberate post-generation alteration of numerical values.

**LEGAL EVIDENTIARY CERTIFICATION (FRE 902(13)/(14) & ISO/IEC 27037):**
This certificate was generated by an automated, self-authenticating digital forensics pipeline. The cryptographic hash computed at acquisition establishes that the target document has not undergone unauthorized alteration since timestamp capture. All algorithms, feature extractors, and weights are version-controlled and reproducible. This certificate is suitable for submission in civil, regulatory, and insurance dispute proceedings.

**DIGITAL SEAL DOCGUARD**
VERIFIED AUDIT